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15 MARCH 1962
SUPERSEDING
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MILITARY SPECIFICATION

PLOTTING BOARD, TACTICAL DISPLAY

PT-396/AS

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be mounted in a military aircraft to receive ground track data in the form of synchro voltages from a ground position computer, and to present a continuous and/or permanent record of the aircraft's track on a chart or plotting sheet. In addition, the equipment shall receive heading data, and present this information on a heading indicator on the control panel of the plotting board.

1.2 Classification - The Plotting Board Tactical Display PT-396/AS covered by this specification shall be of one type and shall consist of a single unit weighing not more than 20 pounds.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-172	Cases; Bases, Mounting and Mounts, Vibration (For Use with Electronic Equipment in Aircraft)
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-P-7788	Plate, Plastic, Lighting
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures, Preproduction and Inspection; for Aircraft Electronic Equipment; Format for
MIL-N-18307	Nomenclature and Nameplate for Airborne Electronic and Associated Equipment

STANDARDS

Military

FSC 6605

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MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS25245	Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single Phase AC, or 115/200 Volt 3-Phase AC, or Combinations

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to failure of more than 5000 hours should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 20 pounds.

3.2.3 Operating Life

3.2.3.1 Operational Stability - The equipment shall operate with optimum performance for 300 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.3.2 Total Operating Life - The equipment shall have a minimum total operating life of 2,000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

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3.2.3.3* Mean Operating Time Between Failures - The equipment shall have a minimum of 300 hours of mean operating time between failures. A failure shall be defined as any malfunctioning which causes the equipment to fail to meet the minimum performance requirements specified for the equipment.

3.2.4 Cabling and Connections

3.2.4.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.4.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.4.3 Input Power Connection - Input power connections shall be in accordance with MS 25245.

3.2.6 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.7 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.8 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.9 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.10 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient ($30^{0} \pm 10^{0}$ C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0 V AC, at 400 ± 0.5 cps 27.5 ± 0.5 V DC

3.2.11 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment, except as modified herein.

3.2.11.1 Temperature - Temperature requirements shall be from -20^{0}C to + 50^{0}C while operating; -20^{0}C to + 60^{0}C while not operating.

3.2.11.2 Altitude - The equipment shall operate continuously at pressure altitudes from sea level to 30,000 feet.

3.2.12 Warm-up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 10 minutes under standard conditions and 30 minutes at extreme service conditions.

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3.2.13* Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts. (See 6.6)

(1) AC Power (Single Phase) 115V, Category "B" 120 VA
(2) DC Power 28V Category "B" 3 Amps Max., 1.5 Amps continuous

3.2.13.1* Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.2.13.2 Overload Protection - Overload protection for the equipment, shall be provided in the equipment. All parts and circuits which are likely to carry an overload due to any failures or poor adjustments, shall have suitable protective devices, or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

3.3 Performance

3.3.1* General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.1.1 Position of Operation - The construction of the equipment shall be such as to permit operation, continuously or intermittently, in any mounting position up to a 45° angle from the horizontal plane.

3.3.2 Accuracy - The accuracy of the equipment shall be determined as the accuracy to which the stylus is positioned, compared to the computed position as determined by the test inputs. The equipment shall be operated from the output of a navigational computer, or a special testing device simulating the output of a navigational computer and shall be operated to produce a series of tracks of suitable length on the tracing surface. The courses, distances and nautical miles per inch settings shall be so chosen as to thoroughly check the accuracy of the equipment.

3.3.2.1 Accuracy Tolerance - Starting with a given tracer carriage position on a given chart or tracing sheet, the position of the tracer carriage (as determined by the stylus) after a continuous period of operation, shall correspond to the position computed from the input test data within an allowable position error circle of radius 1/16 inch. The center of the error circle shall be located at the computed position.

3.3.2.2 Accuracy of Heading Indication - The heading indicator shall be capable of repeating the heading data supplied by a navigational computer, or special heading simulating device, within 1°.

3.3.3* Tracking Speeds - The equipment shall operate at any tracking speed between 0 and 400 knots when set at any scale, except the 0.5 mile/inch scale which operates with a maximum speed of 200 knots.

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3.3.4* Indication

3.3.4.1* Aircraft's Track - The plotting board shall make a trace of the aircraft's own track on a chart or plotting sheet, by means of a movable carriage and a movable stylus. The stylus shall consist of a standard 0.046 diameter pencil lead of suitable grade.

3.3.4.2* Plot Light - It shall be possible to observe the aircraft's instantaneous position, during tracking, as a spot of light with cross-hairs, projected through the window of the unit cover.

3.3.4.3* Distinctive Mark - The equipment shall be capable of indicating the position of the aircraft with respect to a position being flown over on the ground. This position shall be indicated by a circle marked on the plotting sheet by the stylus.

3.4 Detail Requirements -

3.4.1 Plotting Board, Tactical Display PT-396/AS - The plotting board shall meet the following requirements:

3.4.1.1 Function - The plotting board shall operate from signals received from a navigational computer. The signals received from two approved autosyns shall be at a scale factor of 0.5 revolution of the synchro rotor per nautical mile, to distance N-S (miles north-south) and E-W (miles east-west) from any reference point. A true heading signal is also supplied at 360° per revolution (by the autosyn in the navigational computer) to a heading indicator on the plotting board control panel.

3.4.1.2 Form Factor and Mounting - The dimensions of the plotting board (not including mounting or external connections) shall not exceed 19 inches in width, 24 inches in length, and 3 1/8 inches in depth. No separate mounting tray shall be provided. The shock mounts shall be attached directly to the base of the plotting board and shall be easily removed.

3.4.1.3 Weight - The total weight of the unit shall not exceed 20 pounds.

3.4.1.4 Contents of the Unit - This unit shall contain circuits and components necessary to fulfill the functions described herein. This includes necessary amplifiers, which shall be attached directly to the base of the plotting board.

3.4.1.4.1 Control Panel - The control panel shall be located along the front edge of the unit. The panel shall have a plastic lighting plate meeting the applicable requirements of MIL-P-7788. Control panel lamps shall be readily replaceable without the need for special tools.

3.4.1.4.2* Tracer Carriage - The tracer carriage shall be a movable platform whose position shall be controlled by the signals received from a navigational computer. The tracer carriage shall carry a stylus (pencil) located so that a continuous trace of the aircraft's own track shall be made on a chart, or plotting sheet. It shall be possible to raise and lock the stylus off the plotting surface for repositioning of the carriage or replacing of the stylus. The stylus carriage shall provide maximum visibility of the trace from the operator's position. Provisions shall be incorporated for braking or locking the carriage mechanism so that it is not free wheeling when the switch is in the STBY position or when no power is applied to the unit.

3.4.1.4.3* Carriage Drive - The tracer carriage shall operate over a plotting area of 15 by 15±1/16 inches. The motion of the carriage within this area shall be determined by N-S and E-W ground track synchro inputs. The carriage shall be driven within the plotting area by suitable shafting, mylar tapes and gearing within the plotting board. The overall gear ratio between N-S and E-W servo units and the carriage shall be varied through use of a scale selector as described in paragraph 3.4.1.5.(6). The contact rail design shall insure against open circuit of the stylus mechanism.

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(a)* Servo Units - The N-S and E-W ground track servo units shall be identical, interchangeable plug-in units and shall be mounted within the control panel section of the equipment with required interconnecting gearing and electrical connections. Each servo unit shall consist of a matching servo motor synchro, control transformer and connecting gearing.

3.4.1.4.4 Stylus (Pencil) Assembly - The stylus assembly shall contain the stylus (pencil), the mark mechanism and the plot light assembly as described below.

(1) Stylus - The stylus shall make a dense trace of the aircraft's own track or mark the aircraft position on the plotting sheet. The stylus shall consist of a replaceable, standard 0.046 inch diameter pencil lead of suitable grade.

(2) Mark Mechanism - The distinctive mark shall be actuated by a rotary solenoid which shall be provided as part of the carriage assembly. When the solenoid is momentarily energized, it shall rotate the eccentrically mounted stylus carrier to make the distinctive mark.

(3) Plot Light Assembly - The plot light provides a spot of light with cross-hairs visible through the plotting surface window. The light and housing shall be part of the stylus assembly and shall provide for easy replacement of the lamp. Power requirements for this light and the mark solenoid shall be supplied through necessary contact rails and brushes in the plotting board.

3.4.1.4.5 Plotting Surface - The top cover of the plotting board, except for the control panel area, shall be a smooth, flat surface suitable for manual plotting purposes. The flat surface shall consist of a transparent window, 1 8 inch thick by not less than 15 inches square, and positioned directly over the plotting area. The transparent window shall conform to specification MIL-P-77 and shall be suitable for marking with a grease pencil. Provisions shall be incorporated for mounting paper on the top surface of the window. The cover shall be hinged to the back of the plotting board to provide access to the tracer carriage assembly and the plotting base. The cover shall be supported in the open position. No portion of the unit (such as controls) shall extend above the plotting surface.

3.4.1.4.6* Plotting Base - The paper clips on the plotting base shall be located so that the paper size is not critical. At least 1/4 inch tolerance shall be allowed for paper size. The base shall also be indexed so that graph paper can be inserted properly.

3.4.1.5* Controls - The equipment shall be provided with the controls and indicators listed below:

	Control or Indicator	Setting	Function
(1)	POWER toggle switch	On Off	main power switch controls AC, DC and synchro signals to the equipment
(2)	TRACK MODE toggle switch	STBY	In this position magnetic clutches are de-energized and no tracking takes place
		TRACK	In this position automatic tracking takes place and the tracing pencil traces the ground track
	track indicator		red light that indicates when toggle switch is in TRACK position

NOTE: Provisions shall be made to remotely start and stop the movement of the tracer carriage specified in paragraph 3.4.1.4.2.

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	Control or Indicator	Setting	Function
(3)	E-W positioning knob	E(east) counter-clockwise rotation W(west) clockwise rotation	manual positioning of east-west components of tracer carriage motion
(4)	N-S positioning knob	N(north) clockwise rotation S(south) counter-clockwise rotation	manual positioning of north-south components of tracer carriage motion
(5)	TRACK ILLUM rotary switch	OFF to 1/2 brilliance in 1 step. Half to full brightness in seven steps	controls tracing area lights. Further clockwise rotation increases brightness
(6)	SCALE MI./IN. rotary switch	1/2, 1, 2, 4, 8, 16	determines the scale (in miles of aircraft track to one inch of plotted line) to which the trace line will be drawn

NOTE: The scale selector mechanism shall provide positive detent action for each scale setting to prevent leaving the gear changer line open inadvertently.

(7)	MARK pushbutton	push	causes the pencil to mark a small circle on the track trace, to indicate the position of the aircraft with respect to a position being flown over on the ground
(8)	HEADING INDICATOR 360° indicator		displays aircraft heading with respect to true-north, as transmitted by navigational computer

(9) Pencil Positioning Controls - Manual positioning of the stylus carriage in the north-south direction shall be controlled by the N-S positioning knob. The safety lock shall be unlatched, the knob shall be depressed and rotated in a clockwise direction to introduce a north component, or rotated counter-clockwise to introduce a south component of stylus motion. The east-west components of stylus motion shall be controlled by the E-W positioning knob on the right hand side of the control panel, whereby east shall be a counter-clockwise rotation, and west shall be a clockwise rotation of the knob.

(a) Manual Correction - It shall be possible to manually set the position of the carriage at any time, including times when the plotting board is operating. This shall be permitted by the action of a microswitch under each pencil positioning control knob. When each knob is depressed to slew, the switch shall operate an electro-magnetic clutch in the corresponding gear train, which shall stop the servo drive and permit the manual input to be introduced without incident. Releasing of the carriage positioning control knob shall cause the servo drive to immediately resume the original track direction and magnitude. Latches shall be provided to prevent accidental depression of the slew controls.

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(b) Slewing Speeds - The N-S and E-W pencil positioning controls shall be capable of moving the stylus over the plotting surface at a rate of at least 2 inches per second.

3.4.1.6 Internal Illumination - The internal lighting system shall be provided to illuminate the chart surface with white lights. This circuit shall be controlled by the TRACK ILLUM switch.

(1)* Red Lighting - As required for dark adapted operation. It shall be possible to change from white to red lighting by a simple service change. The change shall consist of a replacement of the clear lamps in the lighting units with red lamps. The red lamps shall be red-stained glass and not surface dyed glass.

(2) Lighting Regulation - The illumination level of the plot area shall be controllable from off to approximately half brilliance in one step and from approximately half to full brilliance in seven additional steps. The pencil trace shall be visible in the top surface of the plotting board when the switch is in position 2. At full voltage, the minimum brilliance shall be not less than 0.25 foot lamberts.

(3) Brightness - Variation in brightness over the entire transparent window area shall not exceed 7:1. The gradient of brightness over any nine square inch area shall not exceed 3:1.

3.4.1.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J301	MS-3106E-18-1S	Power input and provision for remote mark switch
J302	MS-3102E22-14B	Input signals from navigational computer

3.4.1.7 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment.

Nomenclature	Function
Computer Group, Navigational AN/ASA-13,AN/ASA-13A, or equivalent	Furnish N-S and E-W ground track and heading data
Automatic Navigator AN/APN-67	Furnish N-S and E-W ground track and heading data
Test Set, Dead Reckoning Tracer AN/ASM-34	To test plotting board

4.* QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

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4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.1.1 Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. This data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. Such data on parts may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Equivalency - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by this government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

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4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction and performance and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor, or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

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Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests:

Quantity of Equipments Offered for Acceptance	Quantity To be Tested
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction and special tests.

(3) Manufacturing reliability test specified in para. 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

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Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory control.

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity, for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - a 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following: When a production sample equipment is submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room Ambient
A. C. voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. voltage	27.0 ± 2.0 volts

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4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

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4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 115 ± 1.0 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 The parentheses (*) - When used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.9. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity or design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 An asterisk immediately following the paragraph number (e.g. 3.2.13*) indicates either the addition of a new paragraph or a change in the paragraph contents in this revision to this specification.

NOTICE - When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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